# SOFTWARE ENGINEERING PROJECT

## TaskAlign
## (Project Proposal)

## BY


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**DEPARTMENT OF COMPUTER ENGINEERING**
**FACULTY OF ENGINEERING**
**KASETSART UNIVERSITY**


**Academic Year 2026**

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# Buravit Yenjit


**This Project Submitted in Partial Fulfillment of the Requirement for Bachelor Degree of Engineering (Software Engineering)**
**Department of Computer Engineering, Faculty of Engineering KASERTSART UNIVERSITY**
**Academic Year 2026**

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(Assoc. Prof.Dr. Punpiti Piamsa-Nga)

# Abstract

Manufacturing facilities, particularly electronic factories with legacy equipment, face significant challenges in optimizing production schedules, often relying on manual planning methods that lead to inefficient resource utilization. This project presents TaskAlign, an task scheduling system designed specifically for electronics manufacturing plants without built-in sensor technology. The proposed system optimizes machine usage and workforce allocation through task sequencing algorithms, enabling factories to undergo digital transformation without requiring expensive equipment upgrades. TaskAlign provides production managers with a user-friendly interface for manual data input and schedule visualization, addressing key challenges such as inefficient task scheduling and machine underutilization. The system analyzes available machines and their capabilities to create optimized production schedules that optimizing user's criteria between operations. By implementing TaskAlign, electronics factories can improve productivity, and enhance scalability when incorporating new product lines. The solution supports factories in their digital transformation journey by providing an accessible approach to production optimization that works with existing machinery.

# Acknowledgement

Put your acknowledgement paragraph here.

Chayakarn Hengsuwan
Buravit Yenjit

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# Chapter 1

# Introduction

## 1.1 Background

Many conventional electronic equipment factories in Thailand still rely on manual scheduling, where production tasks are planned based on human experience rather than optimized strategies. This traditional approach often leads to inefficiencies in resource utilization and production planning. As industries move toward digital transformation, electronic factories face challenges in modernizing their workflows. Our project, TaskAlign, seeks to address these challenges by implementing a job scheduling system that optimizes user-specified criteria, such as production time, machine utilization, and other relevant factors. By incorporating detailed machine-specific data and product specifications, including manufacturing steps and required components, the system generates the most efficient production plan based on the specified parameters. This approach ensures the optimal utilization of resources in the manufacturing process.

## 1.2 Problem Statement

Designing the production scheduling in traditional factories with non-digital machinery usually relies on the experience of the managers rather than automatic optimized scheduling strategies. Without data-driven decision-making, task allocation may not be efficient, leading to extended machine idle times, production bottlenecks, and suboptimal workforce utilization. These inefficiencies increase operational costs and reduce overall productivity. This challenge becomes even more critical as factories scale operations or introduce new product lines, where traditional experience-based scheduling struggles to adapt to complexity. Our project, TaskAlign, provides a solution by allowing factories to input production data into a web application. The system subsequently optimizes job scheduling based on the user's specified criteria, taking into account machine specifications and product assembly requirements, ensuring improved efficiency without requiring costly equipment upgrades.

## 1.3 Solution Overview

TaskAlign is a task scheduling system designed for conventional electronic equipment factories that rely on manual production management. The system assists factory managers and operators in designing job schedules that optimize user-specified optimization criteria, such as production time, machine utilization and workforce allocation, while taking into account constraints like machine cool-down time, energy consumption, and production dependencies. A key feature of TaskAlign is its data encoding system, which transforms user inputs (such as machine specifications, worker assignments, product requirements, and dependencies) into a representation of the factory's workflow that is suitable for job scheduling algorithms. The system then applies optimization algorithms to generate an optimal production schedule that maximizes machine utilization and efficient workforce allocation while respecting constraints like machine cool-down time and production dependencies. Users can view and interact with the resulting schedules through Gantt charts and timeline visualizations, allowing them to make manual adjustments to production plans when necessary.

### 1.3.1 Features

1. **Smart Task Scheduling (Optimization-Based)**
   - Users input machine data, including:
     - Machine name
     - Machine type
     - Machine capabilities

  - Number of fixed workers assigned to operate the machine
  - Maximum operation time
  - Cooldown time
  - Defect rate
  - Production rate
  - Status of machine (Active, Maintenance, Inactive)
  - General product details (Name, Model, Description)
  - Product components:
    * ID (Might be auto generated)
    * Component name
    * Component steps: Steps in assembling this components (Additional details can be added in the "Assembly Process")
    * Total Production time per component production time
    * Required machines
    * Prerequisite components
    * Component quantity
  - Machine capabilities
  - Number of fixed workers assigned to operate the machine
  - Maximum operation time
  - Cooldown time
  - Defect rate
  - Production rate
  - Status of machine (Active, Maintenance, Inactive)
- The system generates an optimal production schedule considering constraints like machine efficiency, energy usage, and human resource availability.

2. **Flexible Scheduling Modes**

- **Fresh Start Mode:** Creates a new schedule for factories starting production from zero.
- **Production Resume Mode:** Users can input completed portions, and the system will recalculate the optimal production plan to complete the remaining work.

3. **Visualized Task Planning (Gantt Chart & Timeline View)**

   - Displays a clear production roadmap, helping managers track daily tasks and machine utilization.
   - Allows users to adjust schedules manually if needed.

### 1.3.2 Optional Features

1. **Cost & Resource Optimization**

   - Users can input energy consumption, carbon emissions, human cost, etc.
   - The system analyzes cost-effectiveness and suggests ways to reduce waste and optimize efficiency.

## 1.4 Target User

### 1.4.1 Target Audience and Challenges

TaskAlign is specifically designed for conventional electronic equipment factories in Thailand facing production scheduling challenges. These facilities typically operate with traditional manufacturing processes and equipment that lack integrated digital systems or sensors for automated tracking. The target users are electronics factories that produce various appliances such as refrigerators, air conditioners, washing machines, and small household electronic devices across multiple production lines.

### Common Challenges

These manufacturing facilities often face common challenges:

- They operate with capable but aging machinery without built-in digital interfaces.
- Production planning relies heavily on manual methods and staff experience.
- They lack real-time data collection capabilities due to the absence of sensors.
- They need to manage multiple assembly lines producing different products simultaneously.
- They require flexible scheduling to accommodate seasonal demand fluctuations.

### Key Stakeholders

The system serves several key stakeholders:

- **Factory Managers** who oversee production operations and seek more efficient methods for scheduling tasks and allocating resources. They need comprehensive views of production timelines and resource utilization.
- **Operators and Technicians** who directly handle machinery and require a user-friendly interface to understand and follow optimized schedules. Since the factories lack automated tracking systems, these users need clear visual representations of the scheduling plan.
- **Factory Owners** looking for cost-effective digital transformation solutions that improve productivity without replacing existing machines. They seek ROI through better resource allocation rather than capital-intensive equipment upgrades.

Users typically possess substantial experience in factory operations with varying levels of technical proficiency. TaskAlign accommodates this diversity through an intuitive interface that provides both basic operational guidance and advanced optimization insights, making digital transformation accessible even to facilities with limited technology infrastructure.

## 1.5 Benefit

TaskAlign offers practical operational improvements for electronics factories. The solution optimizes production schedules according to user-specified criteria, which may include minimizing overall production time, reducing operational costs, or improving energy efficiency depending on the user's criteria. By coordinating machine operations in a logical sequence and efficiently allocating workforce, the system ensures better utilization of existing equipment. The flexibility of TaskAlign allows factories to choose which factors to prioritize in their production planning, adapting to changing business needs or seasonal requirements. Most importantly, TaskAlign enables digital transformation through better process optimization rather than equipment replacement, making production improvements accessible and cost-effective for traditional factories that aren't ready for comprehensive automation upgrades.

## 1.6 Terminology

### 1.6.1 Optimization Algorithm

A computational method used to find the most efficient way to **schedule tasks, allocate resources, and reduce downtime** in the factory based on input constraints (e.g., machine limits, production steps).

### 1.6.2 Task Scheduling

The process of **assigning tasks to machines and workers** in an optimal order, ensuring efficient production while considering machine availability, processing time, and dependencies.

### 1.6.3 Gantt Chart

A visual representation of the production schedule, showing **when each machine is used, what tasks are performed, and how long each step takes**.

### 1.6.4 MRP (I)

Material requirements planning (MRP) is a system for calculating the materials and components needed to manufacture a product. It consists of three primary steps: taking inventory of the materials and components on hand, identifying which additional ones are needed and then scheduling their production or purchase. [1]

### 1.6.5 MRP (II)

Manufacturing resource planning (MRP II) is an integrated information system used by businesses. It evolved from earlier materials requirement planning (MRP) systems by including the integration of additional data, such as employee and financial needs. The system is designed to centralize, integrate, and process information for effective decision-making in scheduling, design engineering, inventory management, and cost control in manufacturing. [2]

### 1.6.6 ERP

Enterprise resource planning (ERP) is a software system that helps organizations streamline their core business processes—including finance, HR, manufacturing, supply chain, sales, and procurement—with a unified view of activity and provides a single source of truth. [3]

### 1.6.7 MES

A manufacturing execution system, or MES, is a comprehensive, dynamic software system that monitors, tracks, documents, and controls the process of manufacturing goods—from raw materials to finished products. Providing a functional layer between enterprise resource planning (ERP) and process control systems, an MES gives decision-makers the data they need to make their plant floor more efficient. [4]

# Chapter 2

## Literature Review and Related Work

## 2.1 Competitor Analysis

TaskAlign is a task scheduling system tailored for conventional electronics equipment factories reliant on manual production management. It empowers factory managers and operators to design optimized job schedules based on user-defined criteria, such as production time, machine utilization, and workforce allocation. The system considers constraints like machine cool-down time, energy consumption, and production dependencies. TaskAlign employs a data encoding system to translate user inputs (e.g. machine specifications, worker assignments, product requirements, dependencies) into a structured factory workflow representation suitable for optimization algorithms. These algorithms generate an optimal production schedule, maximizing machine utilization and workforce allocation while respecting constraints. Users can view and adjust schedules via Gantt charts and timelines, allowing for manual refinement of production plans. While TaskAlign does not provide real-time schedule adjustments, its optimization-based scheduling and customizable workflows offer a powerful solution for electronics manufacturers.

### 2.1.1 Competitor Overview

**Katana MRP**: Katana MRP is a cloud-based manufacturing ERP designed for small to medium-sized businesses. Its strengths lie in sales order fulfillment and inventory management, featuring e-commerce integration and real-time cost tracking. Katana MRP also provides manufacturing analytics and ERP and MES integrations. However, its manual scheduling approach and lack of drag-and-drop capabilities may limit its

| | TaskAlign | Katana MRP | Jobman | SchedulePro |
|---|---|---|---|---|
| **Target Audience** | Electronics Manufacturers | Small to Medium Businesses | Manufacturers and Cabinetmakers | Batch and semi-continuous manufacturing |
| **Platform** | Web Application | Cloud-Based | Web-Based ERP (Enterprise Resource Planning) | Web and On-Premise |
| **Customization options** | Customizable Workflows | Highly Customizable | Customizable Workflows | Limited Customization |
| **Drag-and-Drop Capabilities** | Yes | No information | Yes | No |
| **Integration Capabilities** | | | | |
| **ERP and MES Integrations (existing enterprise system)** | Potential for ERP Integration but not for MES due to lack of real-time tracking | Integrates with E-commerce Tools | Integrates with Production Planning | Limited Integration |
| **Scheduling Optimization** | | | | |
| **AI-Driven Scheduling** | Optimization-Based Scheduling and can also be manually adjustable | Manual Scheduling | Automated Scheduling | Manual Scheduling |
| **Bottleneck Identification and Resolution** | | | | |
| **Predictive Analytics** | No Predictive Analytics | Comes with manufacturing analytics | No Predictive Analytics | Statistically analytics |
| **Cost and Resource Management** | | | | |
| **Cost Tracking** | Cost Summary | Real-Time Cost Tracking | Real-Time Reporting | Estimated Cost Tracking |
| **Resource Allocation** | Resource Allocation | Resource Management | Resource Allocation | Resource Management |

Table 2.1: Competitor Analysis of TaskAlign

suitability for manufacturers requiring more dynamic production control [5].

**Jobman**: Jobman is a web-based ERP solution designed for manufacturers and cabinetmakers. It includes production planning and real-time cost reporting. While Jobman offers automated scheduling and customizable workflows, it lacks predictive analytics and has limited integration capabilities, which may not fully address the needs of electronics manufacturers with complex workflows and existing enterprise systems [6].

**SchedulePro**: SchedulePro is a scheduling software aimed at batch and semi-continuous manufacturing, offering a simple design and on-premise deployment. It provides statistical analytics and resource management features to improve production efficiency. However, its manual scheduling approach, limited customization options, and lack of integration capabilities may not meet the flexibility and connectivity needs of electronics manufacturers looking for comprehensive scheduling optimization [7].

### 2.1.2 How TaskAlign Addresses Competitor Shortcomings

TaskAlign distinguishes itself by catering specifically to the electronics manufacturing sector, offering optimization-based scheduling and cost summaries. Unlike Katana MRP, TaskAlign prioritizes resource allocation optimization. While Jobman provides automated scheduling, TaskAlign offers customizable scheduling tailored to electronics production, including drag-and-drop capabilities for further optimization. In contrast to SchedulePro's limited customization, TaskAlign provides highly adaptable workflows, enabling electronics manufacturers to tailor the system to their unique processes. Although TaskAlign does not offer real-time schedule adjustments, it has the potential to integrate with ERP systems, enhancing overall production management.

## 2.2 Literature Review

Job scheduling optimization is a critical aspect of manufacturing systems, particularly in environments with complex constraints and dynamic requirements. This section reviews three key studies that contribute to advancements in this field.

### 2.2.1 Optimization Model for Production Scheduling Considering Preventive Maintenance

In the dynamic environment of Industry 4.0, [8] proposed a mathematical optimization model for production scheduling that integrates preventive maintenance activities. The model accounts for real-life constraints such as machine downtime, scrap/rework, availability of raw materials, legal overtime limits, and transit times. It aims to create valid production schedules by distributing production orders across available machines while considering constraints like machine capacity, productivity norms, and planned maintenance. The model also allows for timely adjustments to handle uncertainties effectively. Verification was conducted through quasi-real and real-life experiments using data from an automotive manufacturer of locking systems. Results demonstrated the model's ability to optimize production schedules while accommodating preventive maintenance requirements.

### 2.2.2 Data-Mining-Based Real-Time Optimization of Job Shop Scheduling

[9] explored real-time optimization techniques for job shop scheduling problems using data-mining approaches. The study focused on the dynamic nature of job shop environments where real-time data is critical for decision-making. By leveraging historical and real-time data, the proposed method identifies patterns and optimizes scheduling decisions dynamically. The approach integrates machine learning algorithms to predict job completion times and adjust schedules accordingly. Experimental results showed improvements in scheduling efficiency, reduced idle times, and enhanced resource utilization compared to traditional methods. This study highlights the importance of integrating data-driven techniques into job shop scheduling optimization.

### 2.2.3 Multi-Objective Optimal Scheduling with Labor Workload Balance

[10] addressed multi-objective scheduling problems by considering labor workload distribution alongside traditional objectives like minimizing makespan or maximizing machine utilization. The proposed model balances workload distribution among workers while optimizing production schedules. It incorporates constraints related to worker availability, skill levels, and fatigue thresholds to ensure equitable workload distribution. The study used metaheuristic algorithms such as Genetic Algorithms (GAs) and Particle Swarm Optimization (PSO) to solve the problem efficiently. Experimental results demonstrated significant improvements in both workload balance and overall production efficiency, making it a promising approach for human-centric manufacturing systems.

### 2.2.4 Conclusion

These studies collectively demonstrate the potential of integrating mathematical models, data-driven techniques, and multi-objective optimization into job scheduling strategies. They provide valuable insights into addressing practical constraints such as preventive maintenance, real-time adaptability, and labor workload balance.

# Chapter 3

# Requirement Analysis

## 3.1 Stakeholder Analysis

### 3.1.1 Primary Target Users (Direct Beneficiaries)

These stakeholders directly interact with TaskAlign and gain the most benefits from its scheduling optimization features.

**Factory Managers** – Oversee production operations and seek efficient scheduling methods to reduce downtime and optimize machine utilization.

**Production Planners** – Responsible for task scheduling and workforce allocation, ensuring production efficiency and minimizing bottlenecks.

**Machine Operators & Technicians** – Directly interact with machinery and rely on TaskAlign for clear, optimized schedules to streamline manual operations.

### 3.1.2 Secondary Target Users (Indirect Beneficiaries)

These stakeholders benefit from TaskAlign's impact but do not use the system as actively as the primary users.

**Factory Owners** – Interested in cost-effective solutions that improve productivity without requiring expensive equipment upgrades.

**IT & System Administrators** – Manage the implementation and integration of TaskAlign with existing factory systems, including ERP solutions.

**Supply Chain & Logistics Teams** – Depend on optimized production schedules for better coordination of raw material procurement and order fulfillment.

**Customers & Clients** – Indirect stakeholders who benefit from reduced lead times, improved production efficiency, and consistent product availability.

TaskAlign addresses the needs of these stakeholders by providing an optimized, data-driven scheduling system tailored to solve small to medium electronic factories that are still reliant on manual scheduling in manufacturing environments.

## 3.2 User Stories

| **Stakeholder** | **User Story** |
|---|---|
| Factory Manager | I struggle to monitor machine usage and workforce allocation efficiently, leading to production delays. I need a way to streamline scheduling and optimize resource utilization. |
| Production Planner | Manually planning production schedules based on experience often leads to inefficiencies. I need a system that can generate optimized schedules based on machine capabilities and product requirements. But I can also adjust it the way I want and the system shows the time that changes. |
| Machine Operator | Unclear scheduling causes confusion and delays in my tasks. I need a structured and optimized task assignment that ensures smooth operations. |
| Factory Owner | Upgrading machinery is expensive, but inefficiencies in scheduling are increasing costs. I need a cost-effective way to improve production efficiency without heavy investment in new equipment. |
| IT & System Administrator | Manually transferring data between scheduling tools and ERP systems is time-consuming and prone to errors. I need a scheduling system that integrates seamlessly with existing enterprise software. |
| Supply Chain & Logistics Team | Unreliable production schedules make it difficult to coordinate material procurement and deliveries. I need accurate scheduling data to ensure smooth supply chain operations. |
| Customer | Production delays result in late deliveries and inconsistent quality. I need manufacturers to have an efficient scheduling system that ensures timely production and consistent product quality. |

Table 3.1: User Stories by Stakeholder

## 3.3 Use Case Diagram

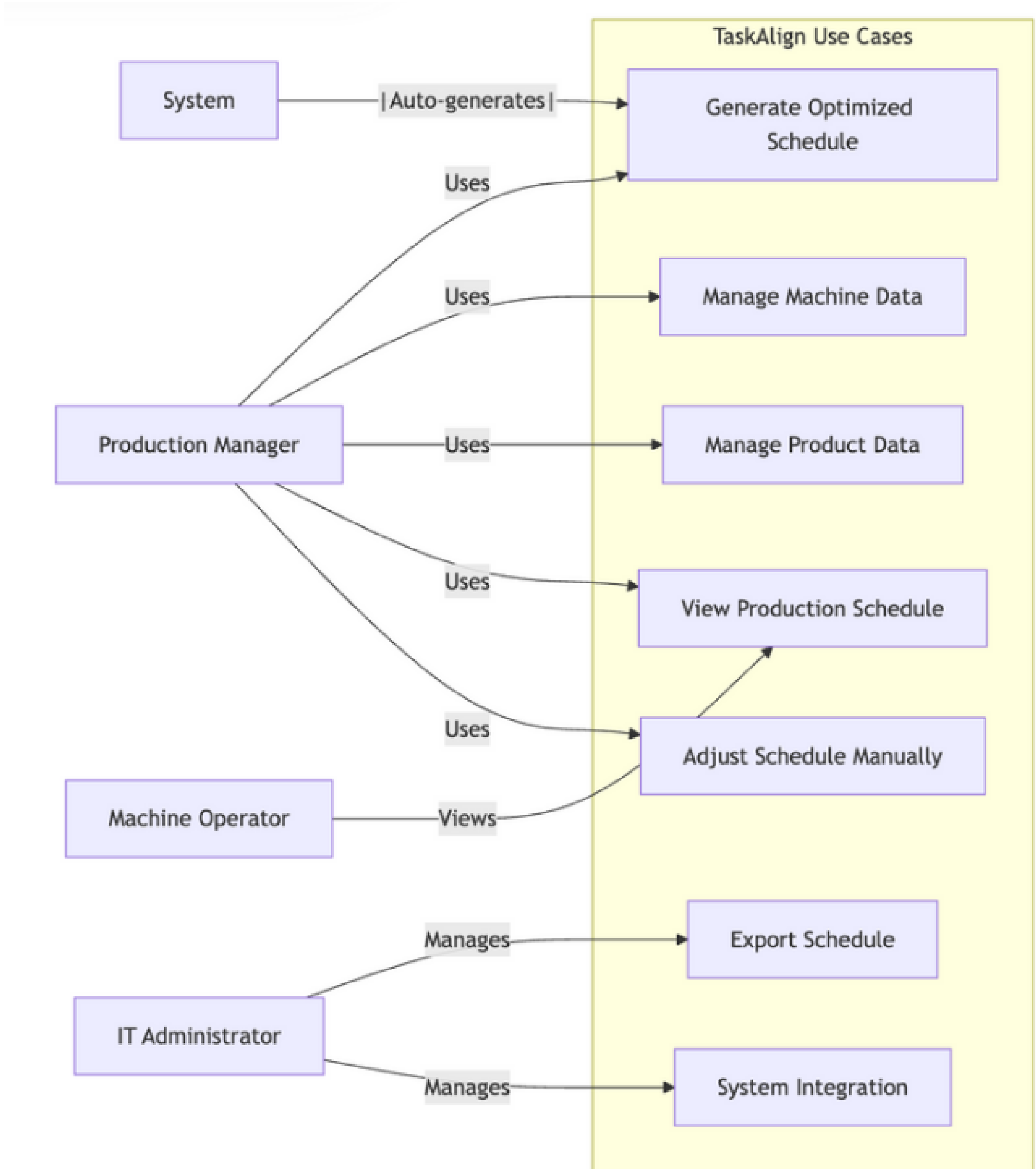


Figure 3.1: Use Case Diagram of TaskAlign

## 3.4 Use Case Model

A use case is a detailed description of how a system interacts with an external entity (such as a user or another system) to accomplish a specific goal. Use cases provide a high-level view of the functionality of a system and help in capturing and documenting its requirements from the perspective of end users.

<TIP: Write use cases for your project here. Make sure to use the appropriate type of use case for each scenario (brief, casual, and fully-dressed use case)./>

## 3.5 User Interface Design

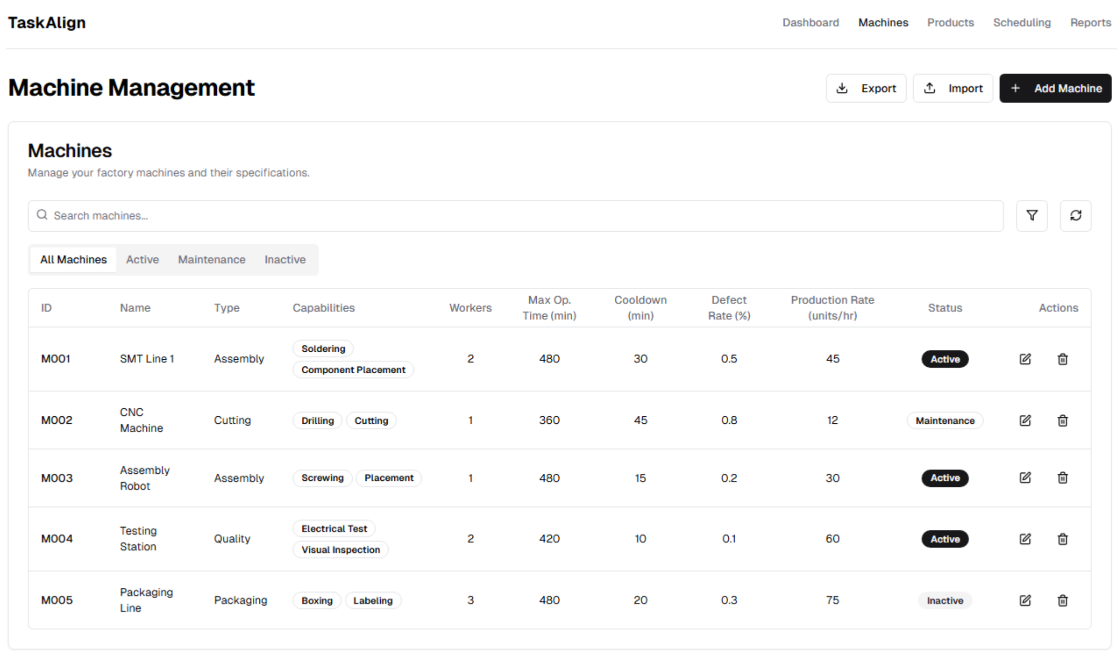


Figure 3.2: Mockup of Machine Management

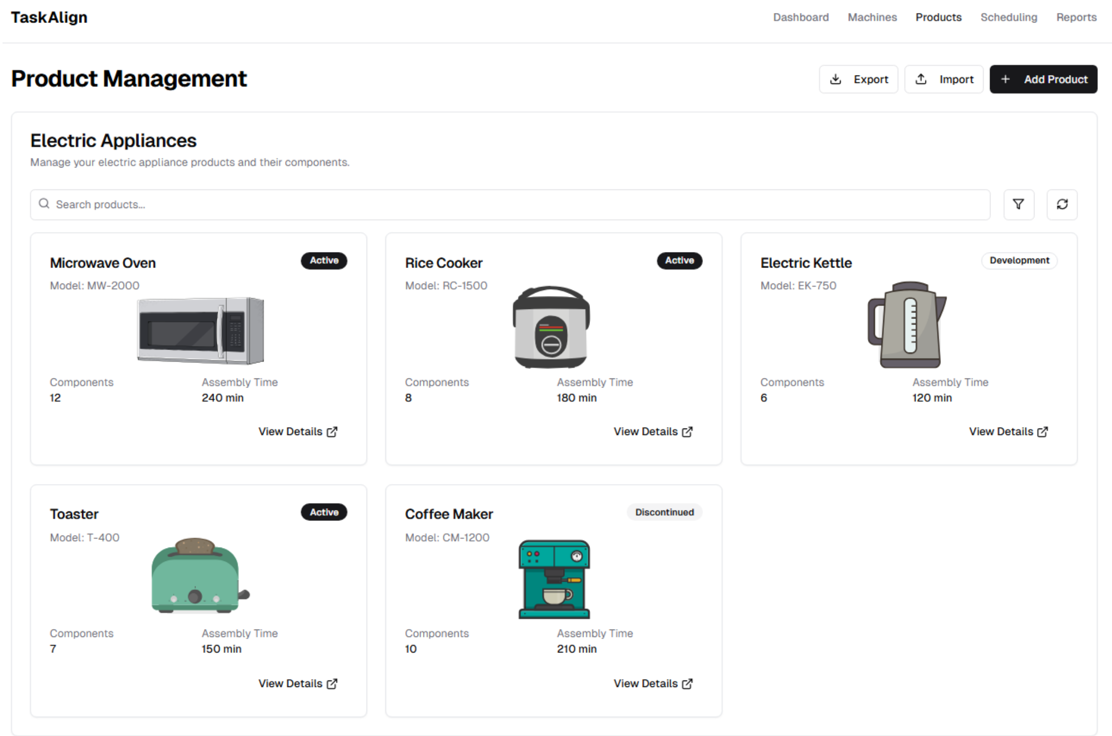


Figure 3.3: Mockup of Product Management

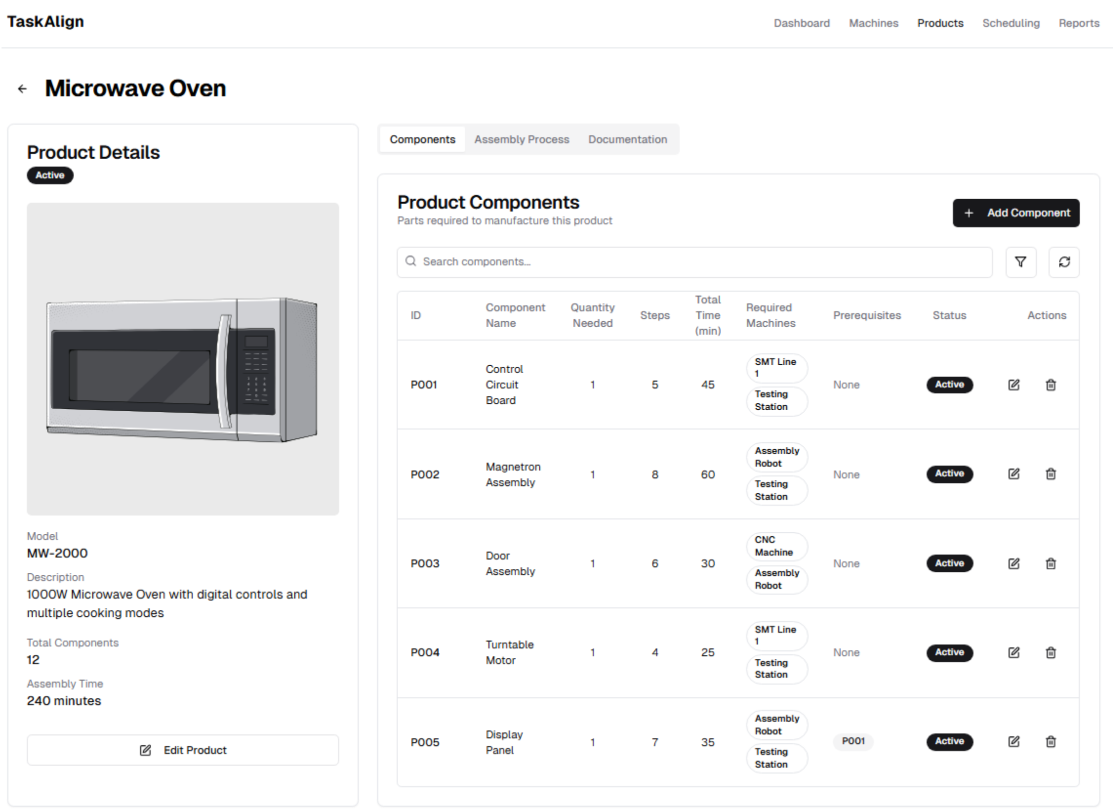


Figure 3.4: Mockup of Product Details, Product Components part

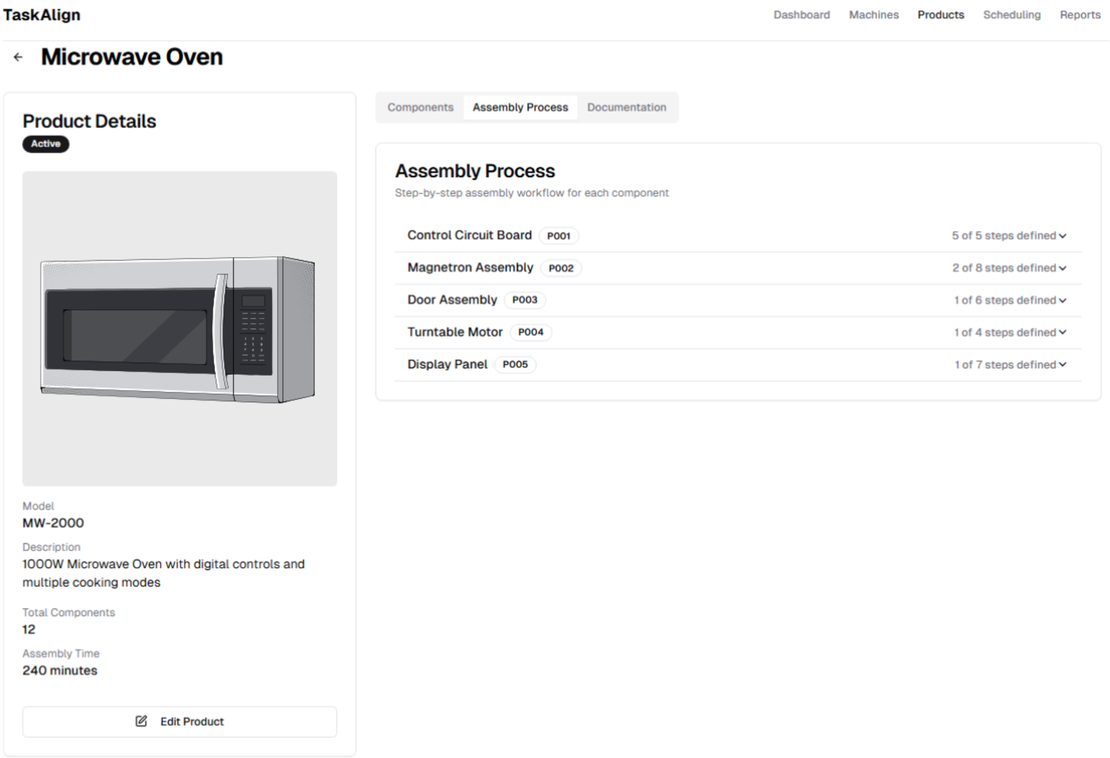


Figure 3.5: Mockup of Product Details, Assembly Process

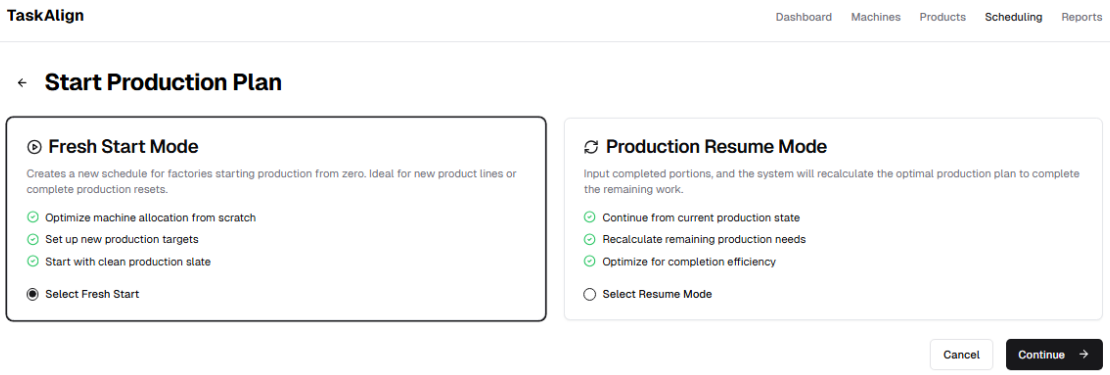


Figure 3.6: Mockup of Schedule Management, Fresh Start Mode

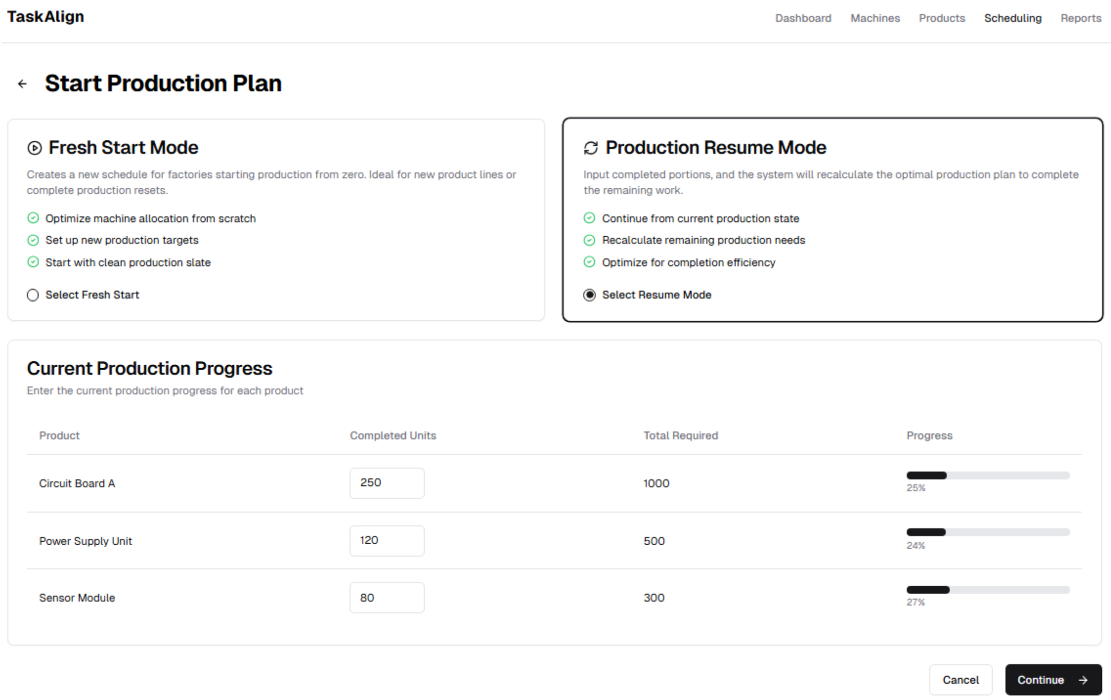


Figure 3.7: Mockup of Schedule Management, Production Resume Mode

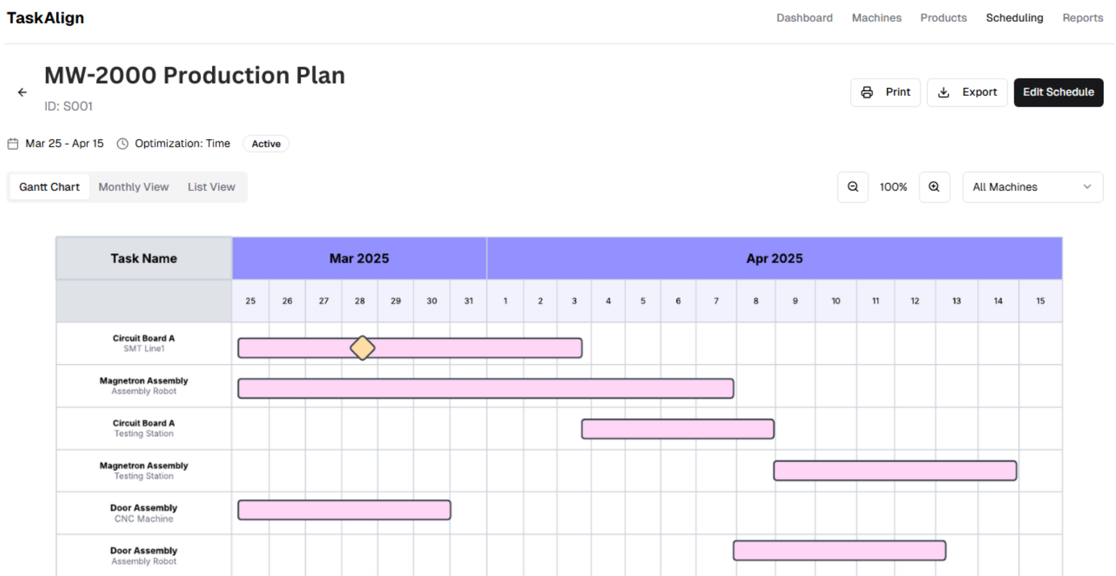


Figure 3.8: Mockup of Gantt Chart in Schedule Management

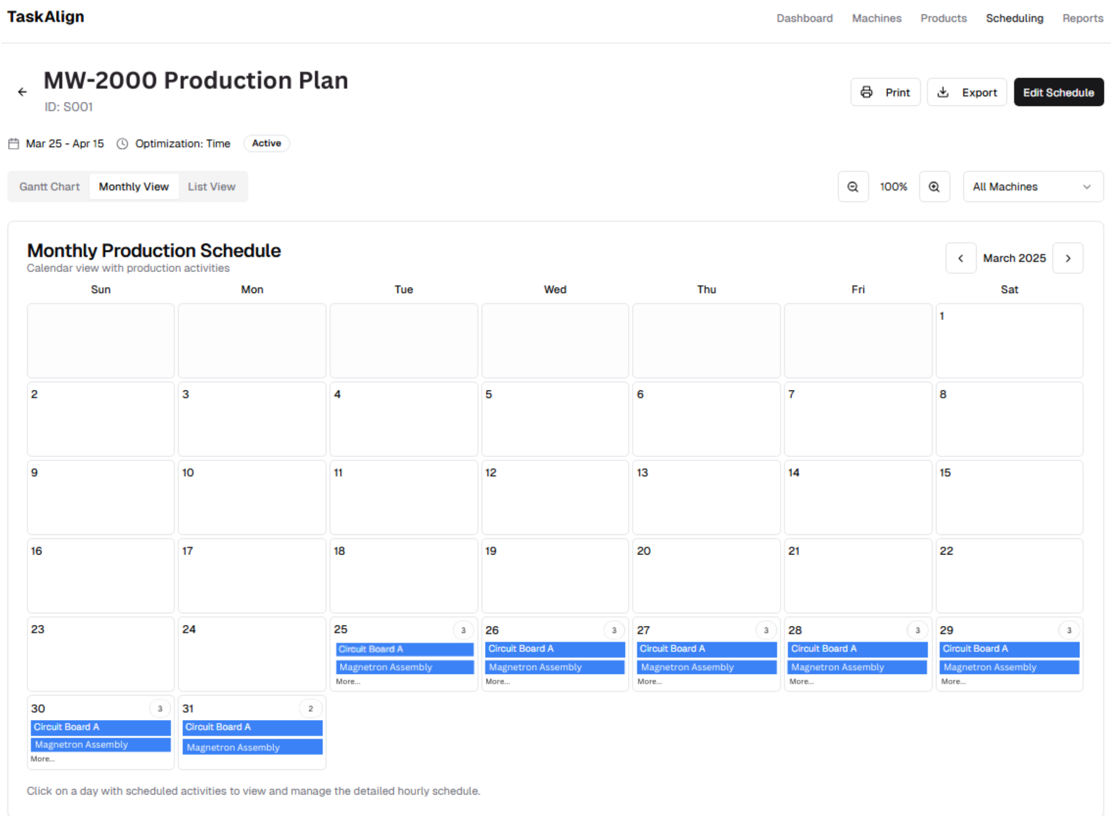


Figure 3.9: Mockup of Monthly View in Schedule Management

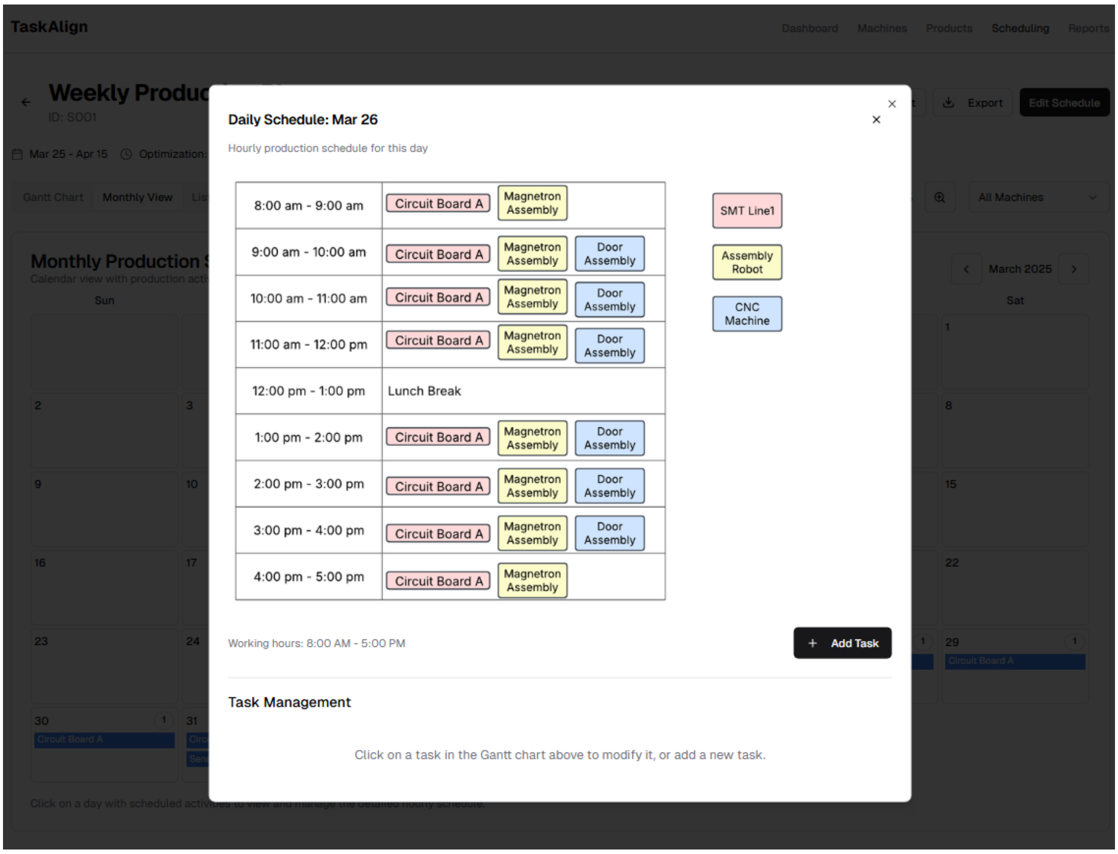


Figure 3.10: Mockup of Daily Schedule in Monthly View

# Chapter 4

## Software Architecture Design

<TIP: Describe how you design your application using Unified Modelling Language (UML). There should be at least two diagrams that describe the software architecture. You may add additional or remove unnecessary diagrams. However, there needs to be a coherency between them at the end./>

### 4.1 Domain Model

<TIP: Describe the business concept of your project. Showcase a domain model that captures the said concept./>

### 4.2 Design Class Diagram

<TIP: Showcase a design class diagram for your project and explain how it works here. You can group classes into packages or layers to communicate your design better./>

### 4.3 Sequence Diagram

<TIP: Sequence diagrams describe how the software runs at run-time. You do not have to create a sequence diagram for every scenario. However, there should be one for all the main ones./>

<ChatGPT: Creating a sequence diagram for every use case is not strictly necessary, but it can be a valuable tool in certain situations. Sequence diagrams are particularly useful for illustrating the interactions

between different components or objects in a system over time, showcasing the flow of messages or actions between them./>

## 4.4 Algorithm

<TIP: Optional, If you are working on a research project that proposes a new algorithm, you can describe your algorithm here. It can be in the form of pseudocode or any diagram that you deem appropriate./>

## 4.5 AI Component

### 4.5.1 Introduction to AI Components

This section outlines the AI components used in TaskAlign, focusing on job shop scheduling and genetic algorithms. These components are tailored to optimize production scheduling for factories without sensor technology, addressing challenges such as inefficient task sequencing, resource underutilization, and extended production times.

### 4.5.2 Job Shop Scheduling

Job shop scheduling is a critical component of TaskAlign, designed to assign tasks to limited resources (machines and workforce) efficiently.

**Input:**

- Machine data: Capabilities, cooldown times, defect rates.
- Workforce data: Number of workers needed for that machine, and machine availability.
- Product data: Manufacturing steps, dependencies.
- Constraints: Task deadlines, machine availability.

**Output:**

Optimized schedules that minimize idle times and bottlenecks.

**Techniques:**

Heuristic methods like First Come First Serve (FCFS) or Earliest Deadline First (EDF) for initial scheduling, followed by optimization using Genetic Algorithms.

### 4.5.3 Genetic Algorithms (GAs)

Genetic Algorithms are utilized in TaskAlign to refine production schedules iteratively by mimicking natural selection processes.

**Input:**

- Initial population of schedules.
- Fitness function evaluating total production time and resource use.
- Constraints like task dependencies and machine cooldown times.

**Output:**

Near-optimal schedules with minimized production time and resource usage.

**Techniques:**

1. Selection of the fittest schedules.
2. Crossover to combine parent schedules into offspring.
3. Mutation for exploring new solutions.
4. Termination upon reaching optimal solutions or iteration limits.

### 4.5.4 Implementation Workflow

The implementation consists of the following steps:

1. Data Collection: Manual input of machine capabilities, workforce details, and product requirements.
2. Initial Schedule Generation: Using heuristic methods like FCFS or EDF.
3. Optimization Using GAs: Iterative refinement of schedules based on fitness evaluation.
4. Visualization: Display optimized schedules using Gantt charts or timelines.
5. Manual Adjustments: Allow users to modify schedules as needed.

### 4.5.5 Optimization Objectives

The system focuses on:

1. Minimizing overall production time by reducing idle times and optimizing task sequencing.
2. Allocating workforce efficiently based on number of workers needed for that machine, and machine availability.
3. Reducing operational costs through optional cost analysis modes.

# Chapter 5

# Software Development

## 5.1 Software Development Methodology

<TIP: Describe your software development methodology in this section. />

## 5.2 Technology Stack

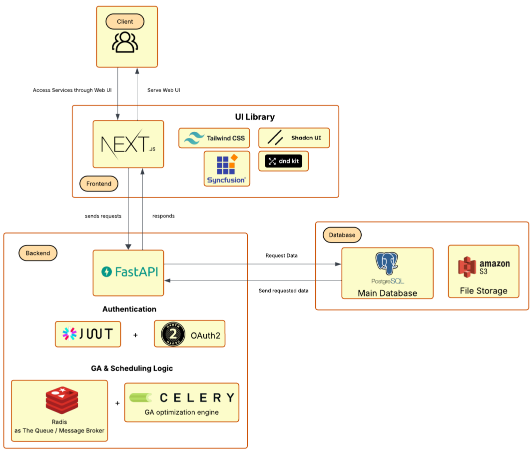


Figure 5.1: TaskAlign's technology stack

## 5.3 Coding Standards

<TIP: Describe your coding standard for this project here. />

## 5.4 Progress Tracking Report

<TIP: Show that you have been working on this project overtime. It can be in the form of a burndown chart or a contribution graph from GitHub./>

# Chapter 6

# Deliverable

## 6.1 Software Solution

<TIP: Share a link to your Github repository. Showcase screenshots of the application and briefly describe each page here. />

## 6.2 Test Report

<TIP: Describe how you test your project. Place a test report here. If you use continuousintegration and deployment (CI/CD) tools, describe your CI/CD method here. />

# Chapter 7

## Conclusion and Discussion

<TIP: Discuss your work here. For example, you can discuss software patterns that you use in this project, software libraries, difficulties encountered during development, or any other topic. />

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# Appendix A

# Appendix A: Example

<TIP: Put additional or supplementary information/data/figures in appendices. />

# Appendix B

# Appendix B: About LaTeX

LaTeX (stylized as LaTeX) is a software system for typesetting documents. LaTeX markup describes the content and layout of the document, as opposed to the formatted text found in WYSIWYG word processors like Google Docs, LibreOffice Writer, and Microsoft Word. The writer uses markup tagging conventions to define the general structure of a document, to stylize text throughout a document (such as bold and italics), and to add citations and cross-references.

LaTeX is widely used in academia for the communication and publication of scientific documents and technical note-taking in many fields, owing partially to its support for complex mathematical notation. It also has a prominent role in the preparation and publication of books and articles that contain complex multilingual materials, such as Arabic and Greek.

Overleaf has also provided a 30-minute guide on how you can get started on using LaTeX. [11]